3.5 Probability

P1

Basic foundation content	Additional foundation content	Higher content only
record, describe and analyse the frequency of outcomes of probability experiments using tables and frequency trees		

Notes: probabilities should be written as fractions, decimals or percentages.

P2

Basic foundation content	Additional foundation content	Higher content only
apply ideas of randomness, fairness and equally likely events to calculate expected outcomes of multiple future experiments		

P3

Basic foundation content	Additional foundation content	Higher content only
relate relative expected frequencies to theoretical probability, using appropriate language and the 0 to 1 probability scale		

P4

Basic foundation content	Additional foundation content	Higher content only
apply the property that the probabilities of an exhaustive set of outcomes sum to 1 apply the property that the probabilities of an exhaustive set of mutually exclusive events sum to 1		

P5

Basic foundation content	Additional foundation content	Higher content only
	understand that empirical unbiased samples tend towards theoretical probability distributions, with increasing sample size	

P6

Basic foundation content	Additional foundation content	Higher content only
enumerate sets and combinations of sets systematically, using tables, grids, Venn diagrams	including using tree diagrams	

P7

Basic foundation content	Additional foundation content	Higher content only
construct theoretical possibility spaces for single and combined experiments with equally likely outcomes and use these to calculate theoretical probabilities		

P8

Basic foundation content	Additional foundation content	Higher content only
	calculate the probability of independent and dependent combined events, including using tree diagrams and other representations, and know the underlying assumptions	

Notes: including knowing when to add and when to multiply two or more probabilities.

P9

Basic foundation content	Additional foundation content	Higher content only
		calculate and interpret conditional probabilities through representation using expected frequencies with two-way tables, tree diagrams and Venn diagrams